

SMART SPEED MONITORING SYSTEM For Road Safety

DR S. SUGANTHI AND REMIGIUS W.

Advances in embedded systems and sensors have enabled efficient, intelligent surveillance solutions that improve safety and control in transportation and automation applications. The system presented here offers a low-cost, efficient method for real-time vehicle speed monitoring using a microcontroller.

When no object or car is detected, the display shows 'No Car Detected.' Once an object is sensed, the system switches to speed measurement mode. Speed is calculated by measuring the time taken for a vehicle to travel between two infrared (IR) sensors placed a fixed distance apart. The timer starts when the vehicle crosses the first sensor and stops when it crosses the second.

This real-time feedback allows authorities or operators to verify whether vehicles are travelling within safe limits. If the calculated speed exceeds a preset threshold (for example, 60km/hr), the buzzer sounds to indicate overspeed. The system is therefore well-suited for areas requiring strict speed control, such as school zones, hospital campuses, and parking areas.

Fig. 1 shows the prototype assembled on a breadboard. The components required to build the system are listed in the Bill of Materials table.

Block diagram

Fig. 2 shows the system block diagram. It consists of two IR sensors, an Arduino board, an LCD display, and a buzzer that together perform object detection and speed

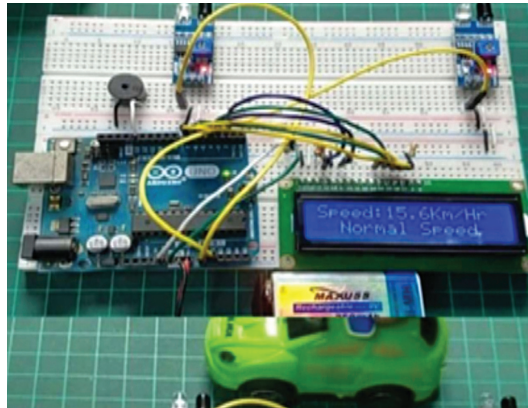


Fig. 1: Authors' prototype

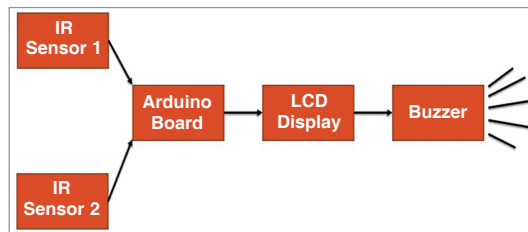


Fig. 2: Block diagram

BILL OF MATERIALS

Components	Quantity
Arduino Uno R3 (MOD1)	1
Buzzer (B1)	1
IR sensor (S1, S2)	2
16x2 LCD display (LCD1)	1
1k potmeter (POT1)	1
Jumper wires	As required

monitoring. The sensors, placed at a fixed distance, detect the movement of an object as it passes between them. The Arduino serves as the main controller, receiving signals from both sensors and calculating speed from the time difference between detections.

The calculated speed and detection status are displayed on the LCD in real time. If the speed exceeds the predefined limit, the buzzer is activated to provide an audible

warning, thereby enhancing road safety.

Circuit and working

Fig. 3 shows the circuit diagram of the smart speed monitoring system. It is built around an Arduino Uno R3 board (MOD1), IR sensors (S1, S2), an LCD display (LCD1), a buzzer (B1), and supporting components. The buzzer sound indicates overspeed.

Two IR sensors are positioned at a known distance for object detection. When a vehicle passes the first sensor, the Arduino records the start time. When the vehicle crosses the second sensor, the end time is recorded. Using the fixed spacing and the measured time interval, the controller calculates the vehicle's speed.

The calculated speed is then sent to the LCD through digital pins. The display shows two outputs: the measured speed (for example, "Speed: xx.xkm/hr") and the detection status ('Car Detected' or 'No Car Detected').

For overspeed monitoring, a buzzer module connected to a digital output pin is controlled through conditional logic. If the calculated speed exceeds a predefined threshold (for example, 50km/hr), the buzzer sounds as an alert. This provides immediate audible feedback in addition to the visual display.

Software

The source code is written in the Arduino IDE, which serves as the core development platform. The program begins by defining pin configurations, setting the IR sensor pins as digital inputs and the LCD and buzzer pins as digital outputs. During